PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Brain Tumor Detection Using AI

submitted in partial fulfilment of the requirements for the award of degree of

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Brain tumor detection using AI"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Punistha**

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Abstract

The advancement of medical imaging technologies, particularly Magnetic Resonance Imaging (MRI), has significantly improved the ability to detect and diagnose neurological disorders. However, the manual identification of brain tumors from MRI scans remains a time-consuming and error-prone process, as it heavily depends on the expertise of radiologists. This limitation can lead to delays in diagnosis, especially in the early stages of tumor development. Therefore, there is a critical need for an automated and reliable detection system.

This paper proposes an AI-based Brain Tumor Detection System that utilizes both machine learning and deep learning approaches to enhance diagnostic efficiency and accuracy. The proposed system processes MRI images through a series of steps, including image resizing, noise reduction, normalization, and feature extraction, to improve data quality and model performance.

To address the problem, multiple models are implemented, including Convolutional Neural Networks (CNN), Random Forest, and Gradient Boosting. The performance of these models is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and Receiver Operating Characteristic (ROC) curve. Comparative analysis demonstrates that deep learning models, particularly CNN, outperform traditional machine learning techniques in terms of accuracy and generalization, owing to their ability to automatically learn complex features from medical images.

The proposed system provides a fast, scalable, and efficient solution for brain tumor detection. By enabling early and accurate diagnosis, it assists healthcare professionals in making timely clinical decisions, ultimately improving patient outcomes and survival rates.

2. Introduction

Medical imaging technologies such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) have significantly improved the ability of doctors to detect and diagnose serious diseases, including brain tumors. These imaging techniques provide detailed visual representations of the brain, enabling clinicians to analyze abnormalities with greater precision. However, despite these advancements, the manual detection of brain tumors from MRI scans remains a challenging and time-intensive task. It requires high expertise and concentration, and even experienced radiologists may encounter difficulties in identifying subtle tumor patterns. This can lead to delays in diagnosis and treatment, which may adversely affect patient outcomes.

To address these challenges, there is a growing need for automated and accurate systems that can assist in the detection of brain tumors. Artificial Intelligence (AI), along with Machine Learning (ML) and Deep Learning (DL), has emerged as a powerful solution in the field of medical image analysis. These technologies enable computers to learn from large datasets, identify complex patterns, and make predictions with high accuracy.

In particular, deep learning techniques such as Convolutional Neural Networks (CNNs) have shown remarkable performance in image classification and analysis tasks. CNNs are capable of automatically extracting relevant features from MRI images, allowing the system to distinguish between normal and tumor-affected brain scans. This reduces the dependency on manual feature engineering and improves the overall efficiency of the diagnostic process.

Although numerous AI-based approaches have been developed for brain tumor detection, it is essential to evaluate and compare different models based on key performance metrics such as accuracy, precision, recall, speed, and reliability. Such comparative analysis helps in identifying the most effective and robust model for real-world medical applications.

Therefore, this study focuses on the development and evaluation of multiple AI-based models for brain tumor detection. By comparing their performance, the research aims to determine the most suitable approach for accurate and efficient diagnosis. The ultimate goal is to support healthcare professionals in making faster and more reliable decisions, thereby improving patient care and treatment outcomes.

4. Methodology

The plan of the Brain tumor detection system follows a systematic machine learning and deep learning pipeline consisting of data collection,preprocessing,feature extraction,model development,Model development,training and testing,evaluation metrics.

A. Data Collection A publicly available MRI brain image dataset is used for this study.The dataset contains labeled images of tumor and non tumor cases.

B.Data Preprocessing The data is preprocessed to improve image quality and model performance.The preprocessing steps includes image resizing,noise reduction,normalization and image enhancement.Libraries used for preprocessing are OpenCV,PIL,NumPy and TensorFlow.

C.Feature Extraction In traditional machine learning models feature extraction techniques are used to capture the important patterns such as texture and shape but in deep learning feature extraction is automatically done by CNN.

D.Model Development Models implemented for this are

1. CNN for tumor classification.
2. Random Forest and Gradient Boosting for comparison.
3. YOLO for tumor detection and localization. 4.U-Net for tumor segmentation.

E.Training and Testing In this step models are first trained by using labeled data and then tested on unseen data.

F.Training and Testing The performance of the model is evaluated on the basis of accuracy,precision,recall,F1-score,ROC curve.

5. Tools and Technologies

The development of the AI-based Brain Tumor Detection System utilizes a combination of programming tools, libraries, and advanced machine learning and deep learning models to ensure accurate and efficient analysis of MRI images.

5.1 Programming Language

- **Python**
  Used for implementing the complete system due to its simplicity and strong support for AI and data science libraries.

5.2 Libraries and Frameworks

- **TensorFlow / Keras**
  Used for building and training deep learning models such as CNN and U-Net
- **Scikit-learn**
  Applied for implementing machine learning models and performance evaluation
- **NumPy and Pandas**
  Used for data preprocessing and numerical operations
- **OpenCV / PIL**
  Used for image processing tasks such as resizing, normalization, and noise reduction
- **Matplotlib / Seaborn**
  Used for visualizing results like accuracy graphs and ROC curves

5.3 Deep Learning Models

- **Convolutional Neural Network (CNN)**
  Used for feature extraction and classification of MRI images
- **YOLO (You Only Look Once)**
  Used for real-time detection and localization of tumor regions
- **U-Net Architecture**
  Used for precise segmentation of tumor areas at pixel level

5.4 Machine Learning Models

- **Random Forest**

  Used for ensemble-based classification
- **Gradient Boosting**

  Used to improve prediction accuracy through sequential learning

5.5 Development Environment

- **Jupyter Notebook / Google Colab**

  Used for model development, training, and experimentation

5.6 Dataset

- MRI brain tumor datasets are used for training and testing
- Images are preprocessed to improve model performance

5.7 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC Curve

6. Major Findings

The proposed system aims to effectively detect brain tumors from MRI images using advanced AI techniques. Preprocessing steps such as image enhancement and normalization are applied to improve image quality and ensure better model performance. A Convolutional Neural Network (CNN) is utilized for accurate classification of MRI images, while the YOLO model is employed to detect and localize tumor regions. Furthermore, the U-Net architecture is used for precise segmentation, enabling clear identification of tumor boundaries.Through comparative analysis, deep learning models demonstrate superior performance over traditional methods in terms of accuracy, reliability, and overall efficiency. This integrated approach provides a robust and effective solution for brain tumor detection.

7. Future Scope

This research demonstrates how AI-based systems are transforming brain tumor detection from a time-consuming manual process into a fast, automated, and efficient solution. By integrating **CNN for classification**, **YOLO for detection**, and **U-Net for segmentation**, the system provides comprehensive diagnostic support.

The combined approach enhances accuracy and enables early detection of tumors, which is crucial for effective treatment. It also significantly reduces analysis time, allowing doctors to make quicker and more informed clinical decisions.

AI in healthcare is not intended to replace medical professionals, but to empower them. By providing reliable, data-driven insights, it supports better diagnosis, reduces workload, and improves overall efficiency. Ultimately, this leads to timely intervention, improved patient care, and higher survival rates.

8. Conclusion

This research is focused on developing an AI-based brain tumor detection system using machine learning and deep learning techniques to faster medical diagnosis.Since manual diagnosis is time consuming and depends heavily on the doctors or the experts,the proposed system aims at providing an automated and efficient solution.The system follows a structured way that include MRI image collection,preprocessing and feature extraction.Deep learning models like CNN ,YOLO,U-Net are used in feature extraction,tumor detection,localization and segmentation. Overall this study highlights the importance of AI in the medical sector and shows how automated brain detection systems can help the doctors in detecting the brain tumor at an early stage which will help them in taking the further steps to cure the brain tumor without wasting any time.

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